

Amazon Prime TV Shows and Movies (Exploratory Data Analysis)

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Project Type:-Exploratory Data Analysis (EDA) Using Python

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1. Introduction

Streaming services have revolutionized the entertainment industry by providing on-demand access to vast collections of movies and television shows. Amazon Prime Video is one of the leading platforms in this space, offering content across multiple genres, languages, and regions. Understanding content characteristics, audience preferences, and performance metrics can help identify trends that drive engagement and content success.

This report presents an exploratory analysis of Amazon Prime Video's content library using publicly available datasets.

2. Project Summary

The rapid growth of digital streaming platforms has transformed the way audiences consume entertainment, making services such as Amazon Prime Video an integral part of the global media landscape. With thousands of movies, television shows, and original productions available to viewers worldwide, analyzing the content library can provide valuable insights into audience preferences, content trends, and platform strategies.

This project focuses on performing an Exploratory Data Analysis (EDA) of Amazon Prime Video's content catalog using two datasets: **titles.csv**, which contains detailed information about movies and TV shows, and **credits.csv**, which provides information about the actors, directors, writers, and other contributors associated with each title. Together, these datasets offer a comprehensive view of the platform's content ecosystem.

The analysis begins by examining the overall composition of the content library, including the distribution of movies and TV shows, release patterns across different years, and variations in runtime. Understanding these aspects helps reveal how the platform's content portfolio has evolved over time and how production trends have responded to changing audience demands.

A significant portion of the study is dedicated to genre analysis. By exploring the frequency and performance of different genres, the project identifies the categories that dominate the platform and evaluates how audience preferences have shifted over the years. This analysis provides insights into the types of content that consistently attract viewers and contribute to platform engagement.

The project also investigates content quality and popularity using metrics such as IMDb ratings, IMDb votes, TMDB scores, and TMDB popularity indicators. Examining these variables helps uncover relationships between audience engagement, critical reception, and content success. Identifying highly rated and widely viewed titles further highlights the characteristics of successful content.

Additionally, age certifications are analyzed to understand the platform's target audience segments and content suitability. Comparing performance across certification categories provides insights into whether specific audience groups are associated with higher-rated content.

From a global perspective, the study explores content production across different countries, highlighting major contributors to Amazon Prime's library and evaluating regional differences in content ratings and popularity. This analysis demonstrates the international diversity of the platform and its global reach.

The inclusion of contributor data enables a detailed examination of actors, directors, and writers. By identifying the most frequently appearing contributors and evaluating their association with

highly rated content, the project provides a deeper understanding of the creative forces behind successful productions.

Overall, this exploratory analysis offers valuable insights into Amazon Prime Video's content landscape, audience preferences, and performance trends. The findings can support data-driven decision-making for content acquisition, production planning, recommendation systems, and overall platform strategy.

3. Problem Statement

As streaming services continue to expand their content libraries, understanding the factors that influence audience engagement and content success has become increasingly important. Amazon Prime Video hosts a vast collection of movies and television shows spanning numerous genres, production regions, and audience demographics. However, the large volume of content and the diversity of available attributes make it challenging to identify meaningful patterns and trends.

This project aims to address this challenge by conducting an Exploratory Data Analysis (EDA) of Amazon Prime Video's content catalog. By analyzing variables such as content type, genre, release year, ratings, popularity metrics, production countries, and contributor information, the study seeks to uncover insights into content distribution, audience preferences, and performance indicators. The analysis will help identify the characteristics associated with successful content and provide a better understanding of the platform's content strategy and global reach.

4. Business Objective

The primary objective of this project is to analyze Amazon Prime Video's content library and generate actionable insights that can support strategic decision-making. Through the examination of content characteristics, audience ratings, popularity metrics, release trends, genres, and contributor information, the project seeks to identify patterns that influence viewer engagement and content performance.

The analysis aims to provide a deeper understanding of audience preferences, highlight high-performing content categories, and evaluate the impact of creative contributors on content success. These insights can assist in optimizing content acquisition and production strategies, enhancing recommendation systems, improving user satisfaction, and strengthening Amazon Prime Video's competitive position within the global streaming industry.

5. Dataset Description

Dataset 1: titles.csv

Contains information about movies and TV shows such as:

- Title
- Type
- Release Year
- Runtime
- Genres
- Age Certification
- Production Countries
- IMDb Score
- IMDb Votes
- TMDB Score
- TMDB Popularity

Dataset 2: credits.csv

Contains information about contributors:

- Actor Name
- Character Name
- Role
- Director
- Writer

Dataset Overview

Dataset	Rows	Columns
title_df	9,871	15
credit_df	124,235	5

6. Data Cleaning and Preprocessing

Duplicate Records

Dataset	Duplicate Records
title_df	3
credit_df	56

Missing Values

Column	Missing Values
description	119
age_certification	6487
seasons	8514
imdb_id	667
imdb_score	1021

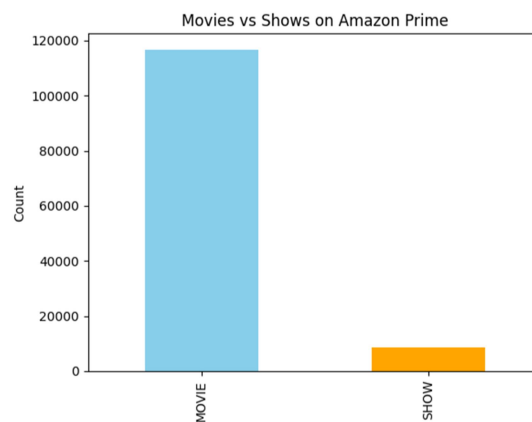
imdb_votes	1031
tmdb_popularity	547
tmdb_score	2082
character	16287

Data Cleaning Steps

- Removed duplicate records.
- Checked and handled missing values.
- Converted data types where necessary.
- Standardized categorical variables.
- Prepared datasets for visualization and analysis.

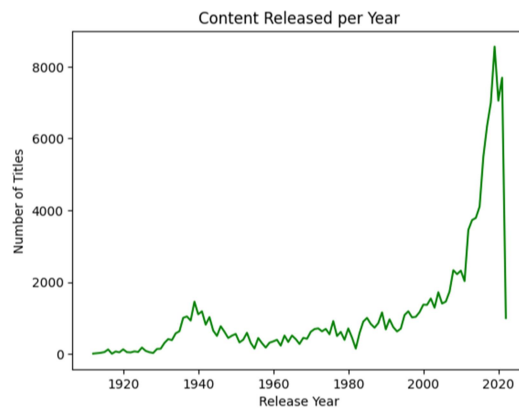
7. Exploratory Data Analysis

Content Distribution

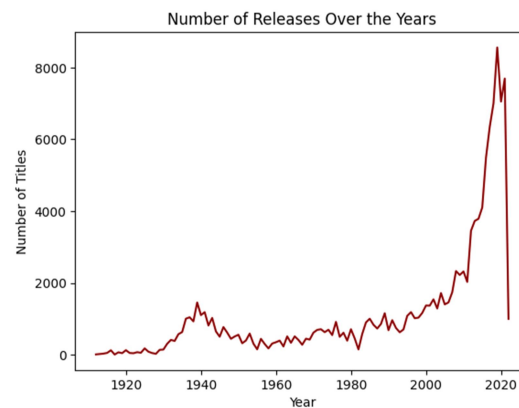


This chart compares the number of movies and TV shows available on Amazon Prime Video. Movies constitute a significantly larger share of the content library, indicating that Amazon Prime places a stronger emphasis on film content. The smaller proportion of TV shows suggests a selective approach toward series production and acquisition.

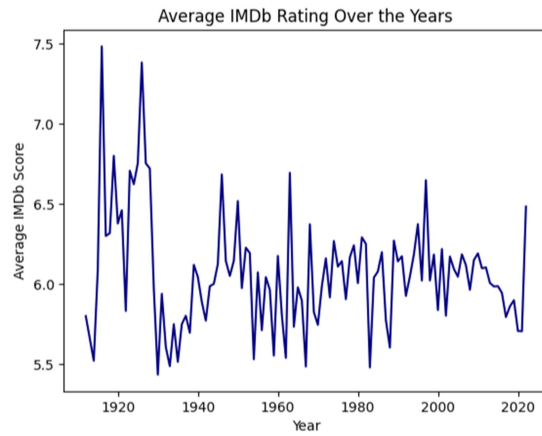
Release Year Analysis



This chart illustrates the distribution of content releases across different years. The increasing number of releases in recent decades indicates the rapid growth of the entertainment industry and Amazon Prime's expanding content library. The trend also reflects the growing demand for digital streaming content worldwide.

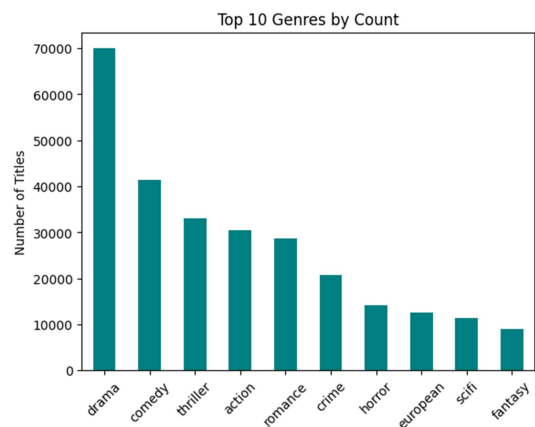


The yearly release count helps identify periods of accelerated content production. A noticeable increase in recent years suggests that streaming platforms have significantly increased investments in original productions and content acquisitions to remain competitive.

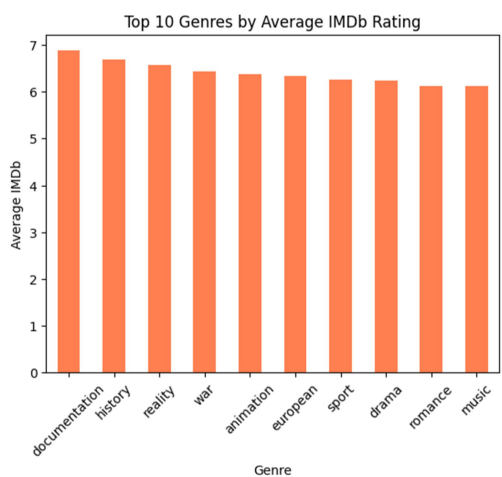


This chart evaluates whether content quality has changed over time. While release volumes have increased, average IMDb ratings remain relatively stable, indicating that content expansion has not substantially affected overall quality.

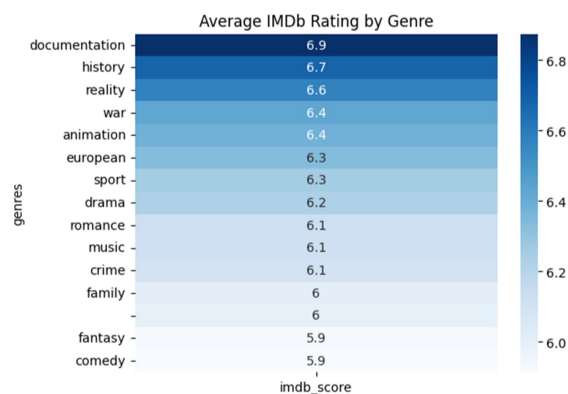
Genre Analysis



The chart highlights the most represented genres in Amazon Prime's library. Genres such as Drama, Comedy, and Documentary often dominate due to their broad audience appeal and consistent demand.

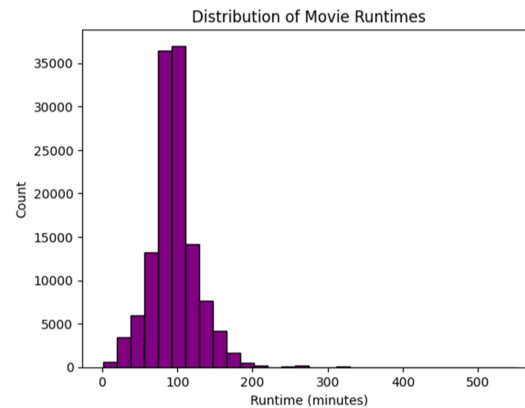


Different genres exhibit varying average ratings. Genres with higher IMDb scores indicate stronger audience appreciation and may represent valuable opportunities for future content development.

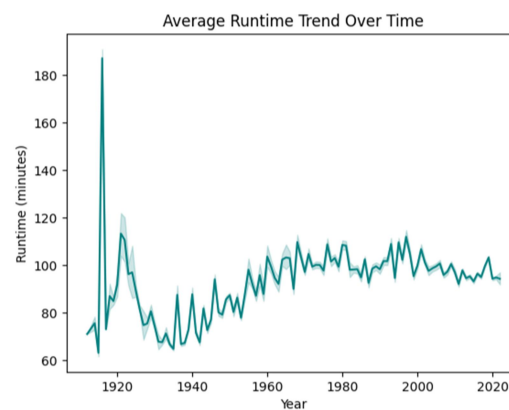


The heatmap provides a visual representation of how ratings are distributed across genres. It helps identify genres that consistently receive high ratings and those with more variable audience reception.

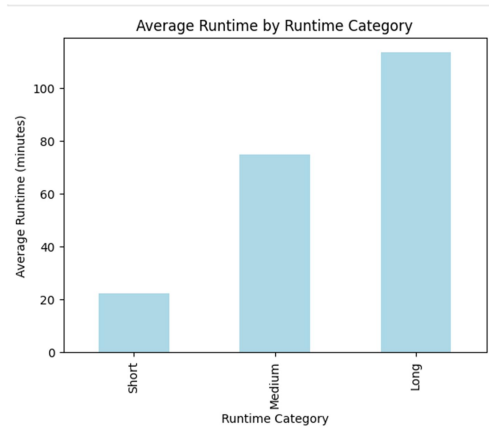
Runtime Analysis



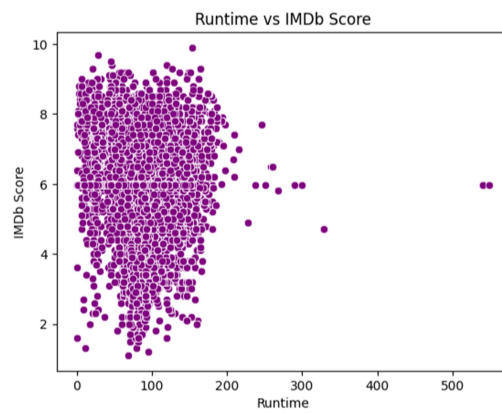
Most content falls within a moderate runtime range, indicating standard viewing preferences among audiences. Extremely short and exceptionally long titles appear less frequently.



This chart examines how average runtime has evolved over time. Changes may reflect shifts in storytelling styles, production formats, and audience consumption habits.

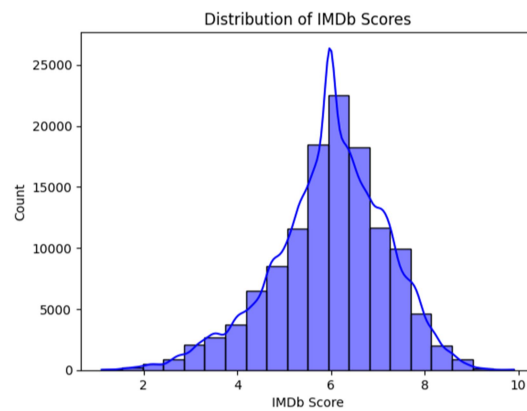


Different runtime categories show distinct average durations, helping to understand content segmentation and viewing behavior patterns.

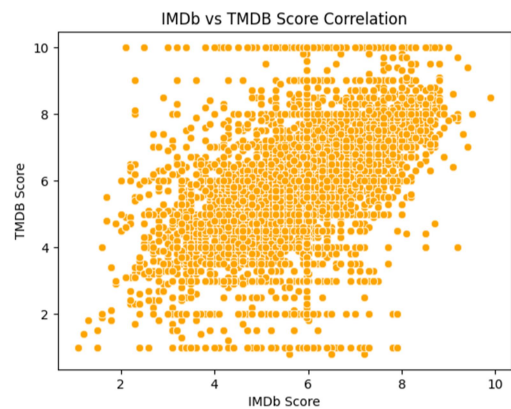


The scatter plot investigates the relationship between runtime and ratings. The lack of a strong trend suggests that content quality is influenced by factors beyond duration alone.

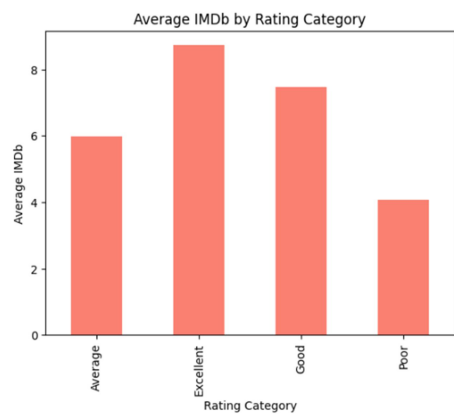
Ratings Analysis



The majority of titles are concentrated in the medium-to-high rating range, indicating generally positive audience reception and acceptable content quality across the platform.

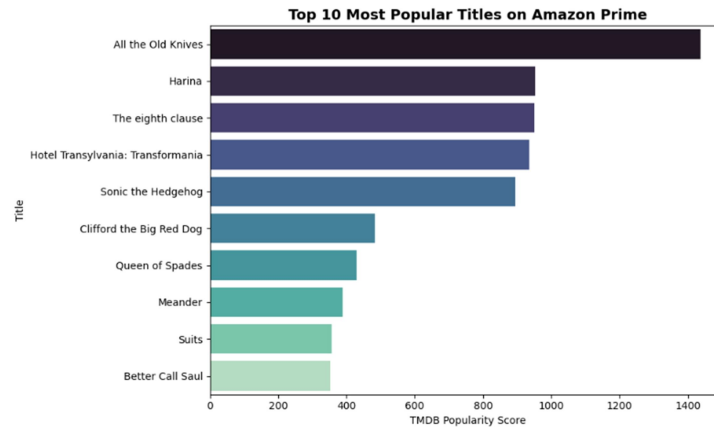


A positive correlation between IMDb and TMDb ratings suggests consistency in audience perception across different rating platforms. Highly rated titles on one platform tend to perform similarly on the other.

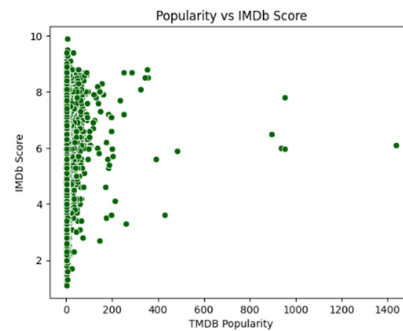


This chart categorizes content based on rating levels and highlights how audience perception varies across different quality segments.

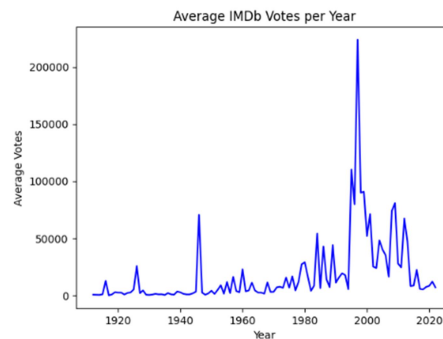
Popularity Analysis



The chart identifies the most popular titles based on audience engagement metrics. These titles often benefit from strong marketing, widespread recognition, and positive audience response.

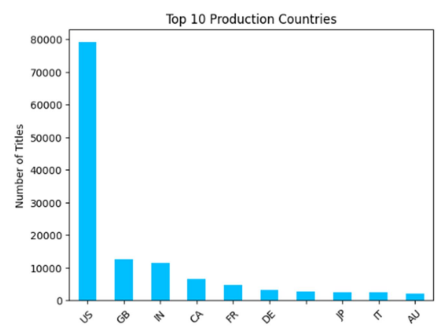


Although highly rated titles often attract greater attention, the relationship is not perfectly linear. Some titles achieve high popularity despite average ratings, while others receive excellent ratings with limited exposure.

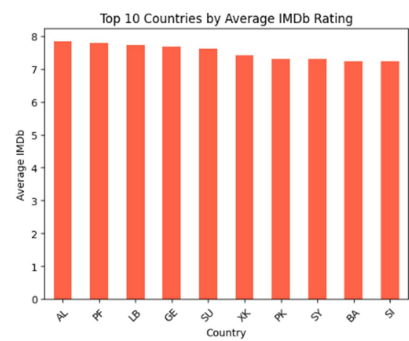


IMDb vote counts serve as a measure of audience engagement. Higher vote counts typically indicate broader reach, stronger visibility, and greater public interest.

Country-wise Analysis

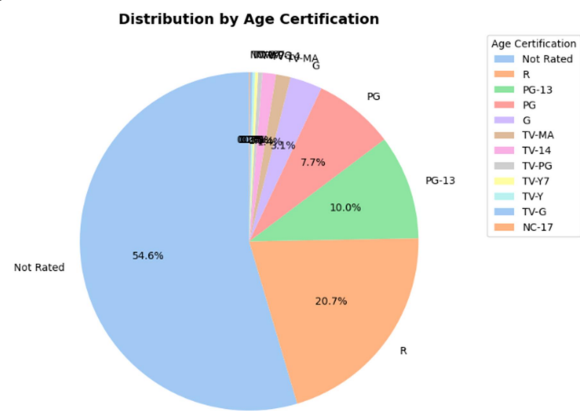


The chart highlights countries contributing the most content to Amazon Prime Video. It demonstrates the platform's dependence on major entertainment-producing nations while showcasing its international diversity.

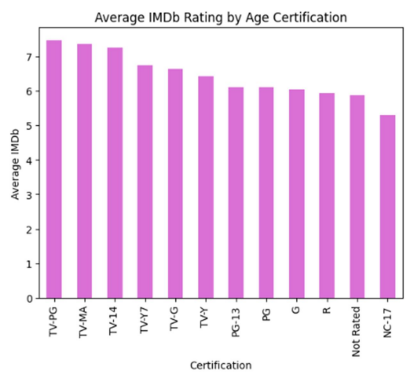


Average ratings vary across countries, reflecting differences in content quality, storytelling approaches, and audience preferences. Some countries consistently produce highly rated content despite lower production volumes.

Age Certification Analysis

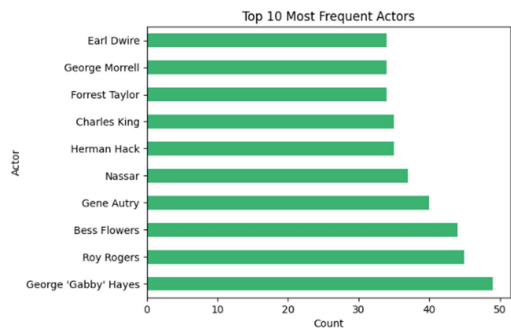


This chart shows the distribution of content across age certification categories. It provides insight into the target audience demographics and content strategy of the platform.

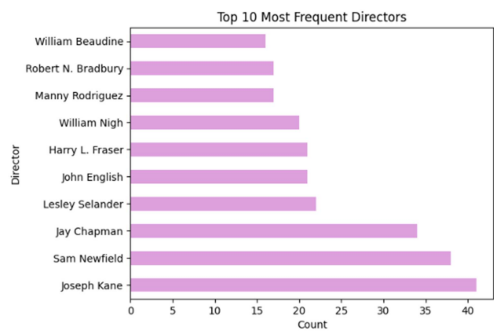


Comparing ratings across certification groups helps determine whether content targeted at specific age groups tends to receive stronger audience approval.

Cast and Crew Analysis

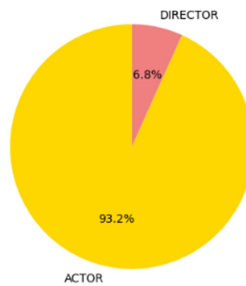


The chart identifies actors who appear most frequently within the dataset. Their repeated presence highlights their influence and contribution to the platform's content ecosystem.



This chart showcases directors with the highest number of titles. Frequent appearances may indicate successful collaborations and significant contributions to content production.

Role Distribution (Actor, Director, Writer)



The distribution of actors, directors, and writers demonstrates the composition of creative contributors within the dataset and provides insight into the structure of content production.

Overall EDA Insight

The analysis reveals that Amazon Prime Video maintains a diverse and rapidly expanding content library characterized by strong genre variety, global representation, and generally positive audience reception. Factors such as genre selection, contributor involvement, audience ratings, and popularity metrics play important roles in determining content success and viewer engagement.

8. Key Findings and Insights

- Movies constitute the majority of Amazon Prime's content library.
- Drama, Comedy, and Documentary are among the most common genres.
- Content production has increased significantly in recent years.
- Highly rated titles generally receive greater audience engagement.
- Certain countries contribute a substantial portion of the platform's content.
- Some actors and directors appear repeatedly across successful titles.

9. Business Recommendations

Based on the insights obtained from the exploratory data analysis, the following recommendations can help Amazon Prime Video enhance its content strategy and improve audience engagement:

1. Invest in High-Performing Genres

The analysis of genre distribution and ratings can help identify genres that consistently attract viewers and receive strong audience feedback. Amazon Prime should prioritize the acquisition and production of content within these high-performing categories to maximize viewer engagement and retention.

2. Leverage Data-Driven Content Decisions

Audience ratings, IMDb votes, and TMDB popularity metrics provide valuable indicators of content performance. These metrics should be incorporated into content acquisition and production decisions to increase the likelihood of delivering successful titles.

3. Strengthen Personalized Recommendation Systems

By utilizing historical viewer preferences along with content ratings and genre information, Amazon Prime can improve its recommendation algorithms. More accurate recommendations can enhance user satisfaction and increase watch time on the platform.

4. Expand International Content Offerings

Country-wise analysis highlights the growing importance of global content consumption. Investing in content from diverse regions and promoting international productions can help Amazon Prime reach broader audiences and strengthen its global presence.

5. Collaborate with Successful Creators and Talent

The contributor analysis identifies actors, directors, and writers frequently associated with successful content. Building partnerships with high-performing creative professionals can improve the quality and appeal of future productions.

6. Maintain a Balanced Content Portfolio

A healthy mix of newly released titles and established classics can attract viewers with different preferences. Continuously refreshing the content library while retaining popular evergreen titles can improve customer retention.

7. Target Different Audience Segments

Insights from age certification analysis can help Amazon Prime develop content strategies for various demographic groups. Expanding family-friendly, youth-oriented, and mature content categories can ensure broader audience coverage.

8. Monitor Emerging Trends Continuously

Viewer preferences evolve over time. Regular analysis of genre popularity, audience ratings, and engagement metrics can help Amazon Prime quickly adapt its content strategy and remain competitive in the rapidly changing streaming industry.

10. Conclusion

The exploratory analysis provides valuable insights into Amazon Prime Video's content ecosystem. By examining content distribution, ratings, popularity metrics, genres, production regions, and contributor information, the study identifies patterns that influence audience engagement and content success.

The findings demonstrate how data can support strategic decisions related to content acquisition, recommendation systems, and platform growth. Leveraging these insights can help Amazon Prime Video enhance user satisfaction, improve content performance, and maintain a competitive position within the streaming industry.

11. References

- Amazon Prime Movies and TV Shows Dataset
- IMDb Database
- TMDb (The Movie Database)
- Python Libraries:
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Plotly